

16762 Project Proposal: Mobile Manipulation for Object Fetch

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March 2026

1 Introduction

In this project, we aim to build an end-to-end system that enables a mobile manipulator to autonomously locate a target object, navigate to it, grasp it, and place it at a designated location.

Given a target object (e.g., “red cup”), the robot must:

- Detect and localize the object in the environment
- Navigate safely to the object
- Execute a stable grasp
- Transport the object to a predefined goal location
- Place the object accurately

2 Proposed Method

2.1 Perception

We use a vision-based object detection model (e.g., YOLO) to detect target objects from RGB images. The detected bounding box is projected into the robot frame using camera calibration to estimate the object position.

2.2 Navigation

We leverage the ROS2 Nav2 stack for autonomous navigation. The robot plans a collision-free path toward the detected object and dynamically updates its trajectory based on sensor feedback.

2.3 Manipulation

We use MoveIt2 for motion planning and control of the robotic arm. A predefined or heuristic grasp pose is generated based on the object location, followed by execution of a pick-and-place sequence.

2.4 System Integration

The full pipeline is organized as:

Detect → Navigate → Align → Grasp → Place